

HIERARCHICAL AGGLOMERATIVE CLUSTERING [HAC]

HAC

- Starts with one cluster , individual item in its own cluster and iteratively merge clusters until all the items belong to one cluster.
- Bottom up approach is followed to merge the clusters together.
- Dendrograms are pictorially used to represent the HAC.

HAC

- Single-nearest distance or single linkage.
- Complete-farthest distance or complete linkage.
- Average-average distance or average linkage.

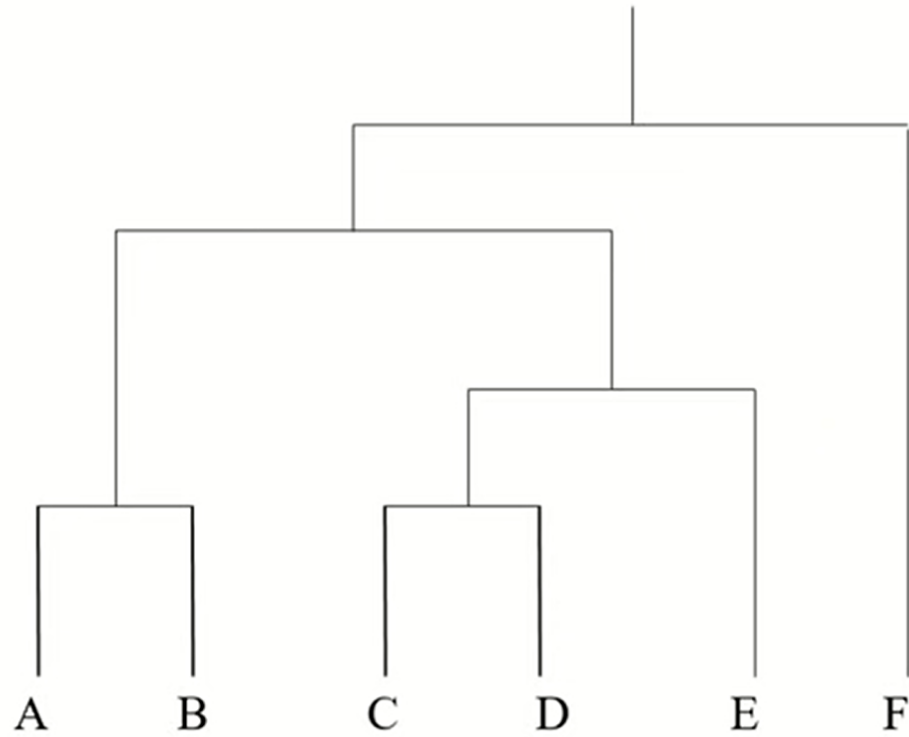
HAC

Single Linkage	This is the distance between the closest members of the two clusters.
Complete Linkage	This is the distance between the members that are farthest apart.
Average Linkage	This method involves looking at the distances between all pairs and averages all of these distances. This is also called Unweighted Pair Group Mean Averaging.

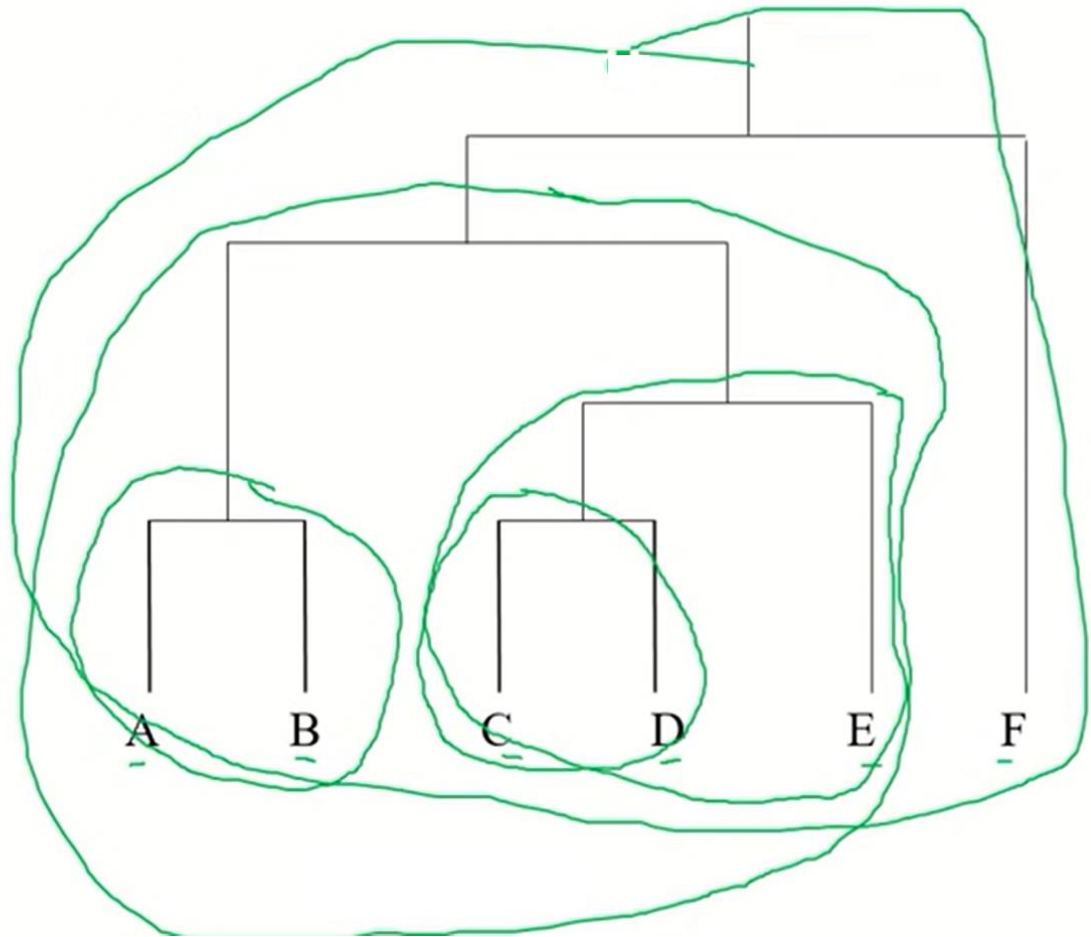
DENDROGRAM

- A tree like structure which represents hierarchical technique.
 - ✓ Leaf- Individual.
 - ✓ Root – One cluster.
- A cluster at level 1, is the merger of its child cluster at level $i + 1$.

DENDROGRAM



DENDROGRAM



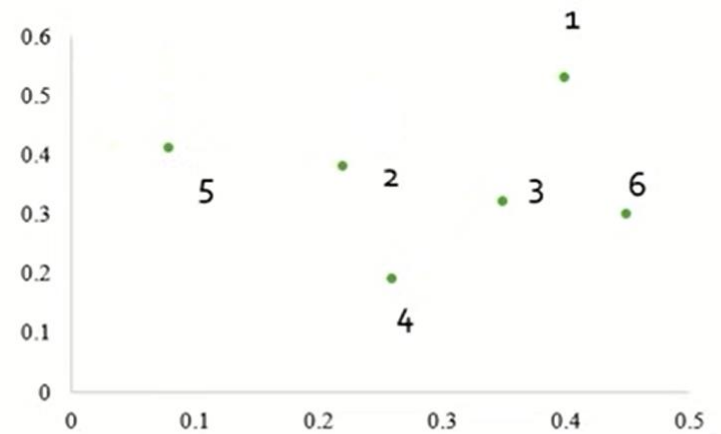
EXAMPLE

- Find the clusters using single link technique. Use Euclidean distance, and draw the dendrogram.

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30

EXAMPLE SINGLE LINK

	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19
P5	0.08	0.41
P6	0.45	0.30



EXAMPLE SINGLE LINK

- Calculate Euclidean distance, create the distance matrix.

$$\text{Distance } [(x,y), (a,b)] = \sqrt{(x-a)^2 + (y-b)^2}$$

$$\text{Distance (P1,P2)} = \sqrt{(0.40 - 0.22)^2 + (0.53 - 0.38)^2}$$

$$(0.40,0.53), (0.22,0.38) = \sqrt{(0.18)^2 + (0.15)^2}$$

$$= \sqrt{0.0324 + 0.0225}$$

$$= \sqrt{0.0549}$$

$$= 0.23$$

EXAMPLE SINGLE LINK

- The distance matrix is

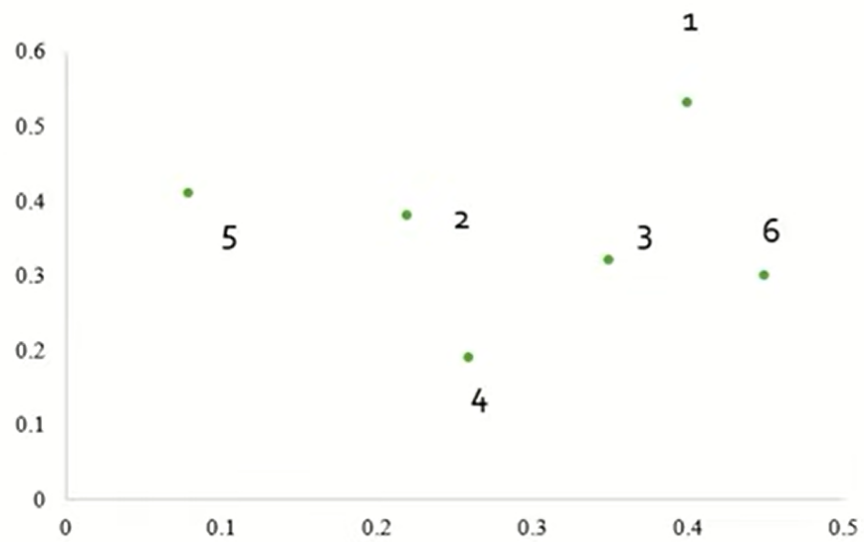
	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.23	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE SINGLE LINK

- The distance matrix is

	P1	P2	P3	P4	P5	P6
P1	0					
P2	0.24	0				
P3	0.22	0.15	0			
P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE SINGLE LINK



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P4	0.37	0.20	0.15	0		
P5	0.34	0.14	0.28	0.29	0	
P6	0.23	0.25	0.11	0.22	0.39	0

EXAMPLE SINGLE LINK

- To update the distance matrix $\text{MIN}[\text{dist}(P3, P6), P1]$
- $\text{MIN}(\text{dist}(P3, P1), (P6, P1))$
 $= \min[(0.22, 0.23)]$
 $= 0.22$
- To update the distance matrix $\text{MIN}[\text{dist}(P3, P6), P2]$
- $\text{MIN}(\text{dist}(P3, P2), (P6, P2))$
 $= \min[(0.15, 0.25)]$
 $= 0.15$

EXAMPLE SINGLE LINK

- To update the distance matrix $\text{MIN}[\text{dist}(P3,P6),P4]$
- $\text{MIN}(\text{dist}(P3,P4), (P6,P4))$
 $= \min[(0.15,0.22)]$
 $= 0.15$
- To update the distance matrix $\text{MIN}[\text{dist}(P3,P6),P5]$
- $\text{MIN}(\text{dist}(P3,P5), (P6,P5))$
 $= \min[(0.28,0.39)]$
 $= 0.28$

EXAMPLE SINGLE LINK

- The updated distance matrix for cluster P3, P6

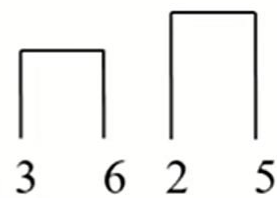
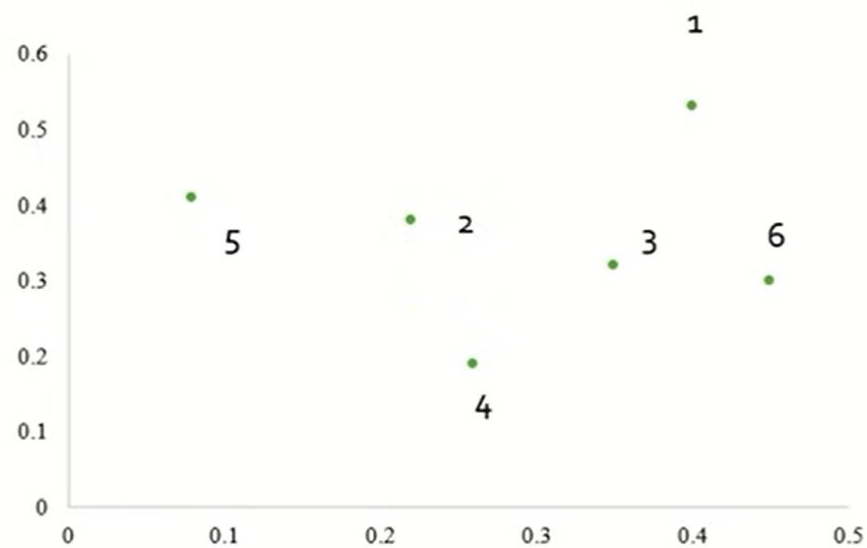
	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.22	0.15	0		
P4	0.37	0.20	0.15	0	
P5	0.34	0.14	0.28	0.29	0

EXAMPLE SINGLE LINK

- The distance matrix fro cluster P2, P5

	P1	P2	P3,P6	P4	P5
P1	0				
P2	0.23	0			
P3,P6	0.22	0.15	0		
P4	0.37	0.20	0.15	0	
P5	0.34	0.14	0.28	0.29	0

EXAMPLE SINGLE LINK



EXAMPLE SINGLE LINK

- To update the distance matrix $\text{MIN}[\text{dist}(P2,P5),P1)]$
- $\text{MIN}[\text{dist}(P2,P1), (P5,P1)]$
 $= \min[(0.23,0.34)]$
 $= 0.23$
- To update the distance matrix $\text{MIN}[\text{dist}(P2,P5),(P3,P6)]$
- $\text{MIN}[\text{dist}(P2,(P3,P6)), (P5,(P3,P6))]$
 $= \min[(0.15,0.28)]$
 $= 0.15$

EXAMPLE SINGLE LINK

- To update the distance matrix $\text{MIN}[\text{dist}(P2,P5),P4]$
- $\text{MIN}[\text{dist}(P2,P4), (P5,P4)]$
 $= \min[(0.20,0.29)]$
 $= 0.20$

EXAMPLE SINGLE LINK

- The updated distance matrix for cluster P2,P5

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

EXAMPLE SINGLE LINK

- The distance matrix is

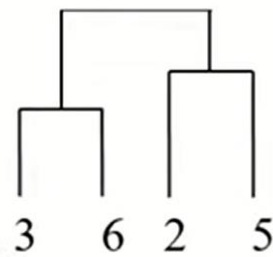
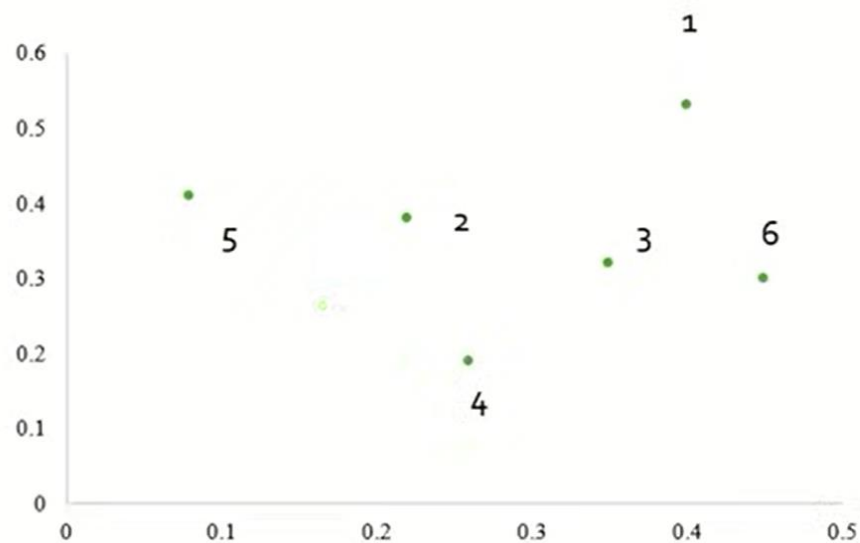
	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

EXAMPLE SINGLE LINK

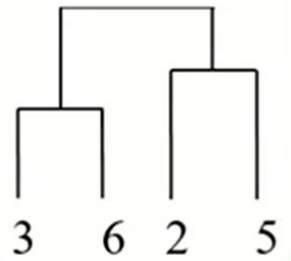
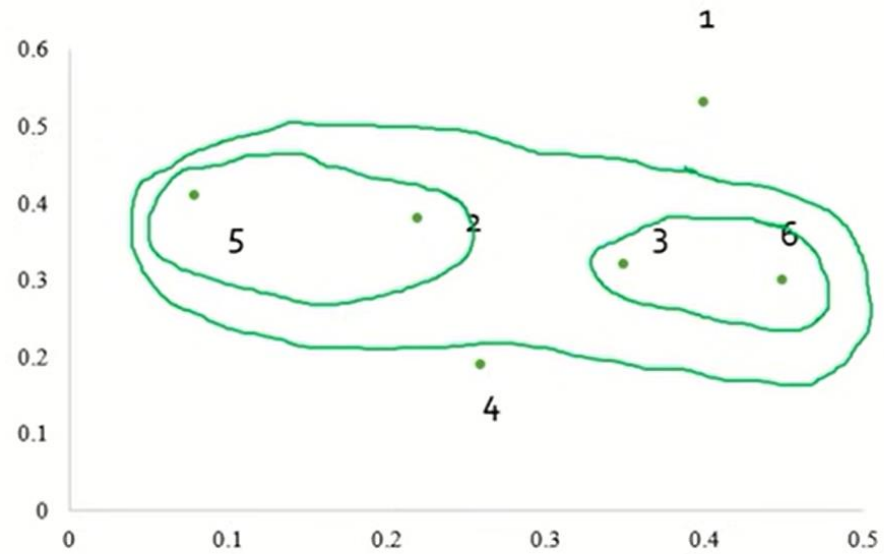
- The distance matrix for cluster (P3,P6), (P2,P5)

	P1	P2,P5	P3,P6	P4
P1	0			
P2,P5	0.23	0		
P3,P6	0.22	0.15	0	
P4	0.37	0.20	0.15	0

EXAMPLE SINGLE LINK



EXAMPLE SINGLE LINK



EXAMPLE SINGLE LINK

- To update the distance matrix $\text{MIN}[\text{dist}((P2,P5),(P3,P6)),P1]$
- $\text{MIN}[\text{dist}((P2,P5),P1), ((P3,P6),P1)]$
 $= \min[(0.23,0.22)]$
 $= 0.22$

EXAMPLE SINGLE LINK

- To update the distance matrix $\text{MIN}[\text{dist}((P2,P5),(P3,P6)),P4]$
- $\text{MIN}[\text{dist}((P2,P5),P4), ((P3,P6),P4)]$
 $= \min[(0.20,0.15)]$
 $= 0.15$

EXAMPLE SINGLE LINK

- The updated distance matrix for cluster P2,P5,P3,P6

	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

EXAMPLE SINGLE LINK

- The distance matrix is

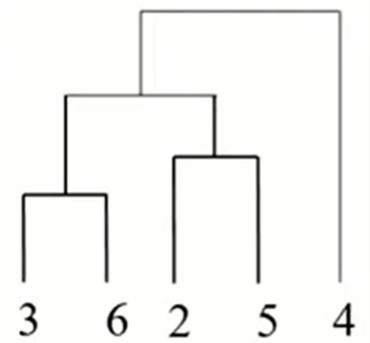
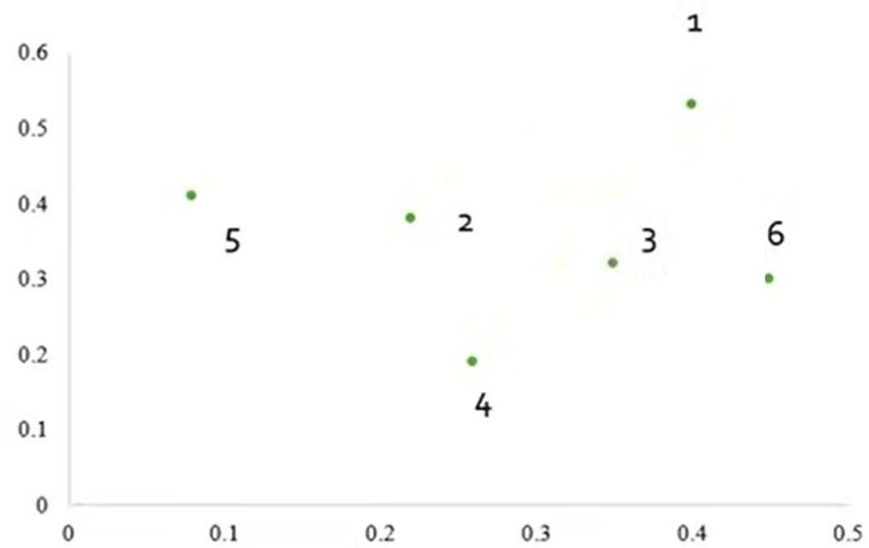
	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

EXAMPLE SINGLE LINK

- The distance matrix is

	P1	P2,P5,P3,P6	P4
P1	0		
P2,P5,P3,P6	0.22	0	
P4	0.37	0.15	0

EXAMPLE SINGLE LINK



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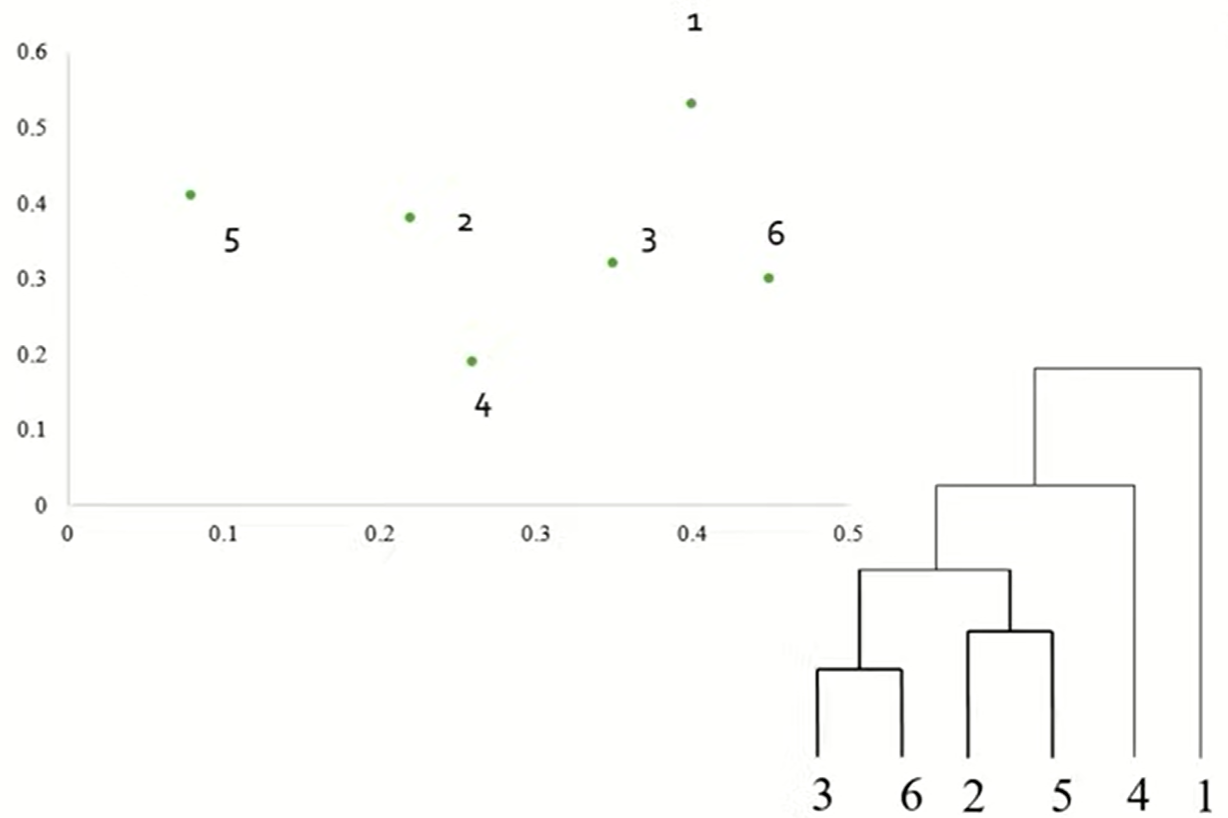
- To update the distance matrix $\text{MIN}[\text{dist}(\text{P2}, \text{P5}, \text{P3}, \text{P6}), \text{P4}]$
- $\text{MIN}[\text{dist}((\text{P2}, \text{P5}, \text{P3}, \text{P6}), \text{P1}), (\text{P4}, \text{P1})]$
 $= \min[(0.22, 0.37)]$
 $= 0.22$

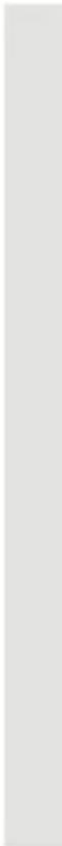
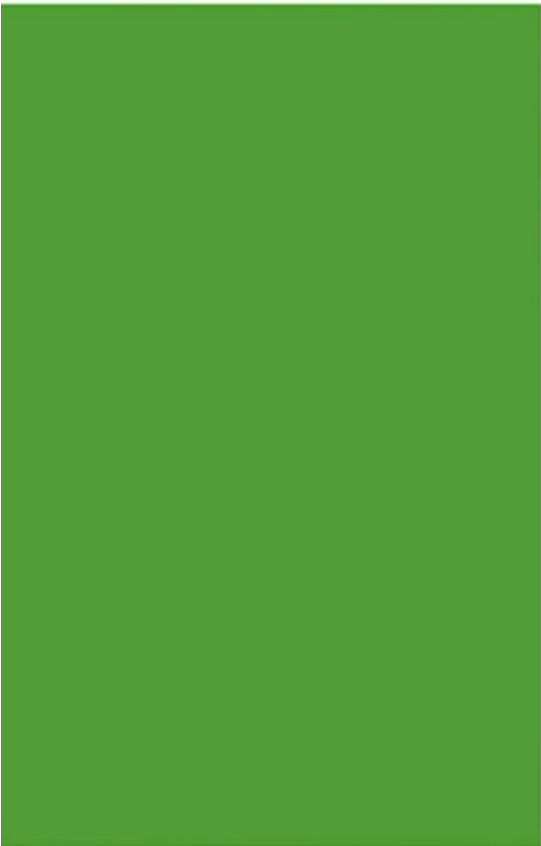
EXAMPLE SINGLE LINK

- The updated distance matrix for cluster P2,P5,P3,P6,P4

	P1	P2,P5,P3,P6,P4
P1	0	
P2,P5,P3,P6,P4	0.22	0

EXAMPLE SINGLE LINK





Thank you